G11 T1 C8 – Student's t: Classroom Activity

Models and simplifying approximation (all pages). Each question supplies a continuous model for a signed deviation: $X = m + aT$, where $T \sim t_\nu$, m is the location parameter (and mean for $\nu > 1$), $a > 0$ is the scale, and $\nu = n - 1$, where n is the number of observations in the sample. For simplicity, the quantity described as an "approximate standard deviation" is used directly as the scale a. This is a teaching approximation: the model's actual standard deviation is $\mathrm{SD}(X) = a\sqrt{\frac{\nu}{\nu-2}}$ for $\nu > 2$, which is larger than a. The approximation is coarser for small ν and improves as ν increases. Use the supplied model without adjusting a. Negative deviations mean below the reference; positive deviations mean above it.

Using the supplied table. In row ν and column α, the positive critical value t^* satisfies $P(T > t^*) = \alpha$. By symmetry, $P(T < -t^*) = \alpha$. Consequently, $P(T \le t^*) = 1 - \alpha$ and $P(T \le -t^*) = \alpha$. Standardize using $t = \frac{x-m}{a}$; back-transform using $x = m + at$. These calculations use the supplied model and its scale a. Use only table entries (no CDF calculator). Table values are rounded to three decimals, so cutoffs computed from them are approximate. Numerical bounds in direct questions retain enough digits to recover the intended table entries. Report probabilities exactly as implied by the table and contextual cutoffs to two decimals. Endpoints have probability zero.

Worked Examples

1. **Direct left tail – arrival error.** The delivery-time deviation from schedule of 8 deliveries, in minutes, has mean 1 and approximate standard deviation 4. Model $X = 1 + 4T$, where $T \sim t_7$. Find the probability that the deviation is below -6.58, i.e., $P(X < -6.58)$.
   Standardize: $t = \frac{-6.58-1}{4} = -1.895$. Row 7, $\alpha = 0.05$: $P(T < -1.895) =$ 0.05. About 5% of deliveries arrive more than 6.58 minutes early.

2. **Direct right tail – cost error.** The cost deviation from budget of 12 batches, in dollars, has mean -10 and approximate standard deviation 20. Model $X = -10 + 20T$, where $T \sim t_{11}$. Find the probability that the deviation exceeds 17.26, i.e., $P(X > 17.26)$.
   Standardize: $t = \frac{17.26+10}{20} = 1.363$. Row 11, $\alpha = 0.10$: $P(T > 1.363) =$ 0.10. About 10% of batches exceed budget by more than \$17.26.

3. **Direct interval – energy error.** The energy-use deviation from forecast over 15 days, in kWh, has mean 5 and approximate standard deviation 10. Model $X = 5 + 10T$, where $T \sim t_{14}$. Find the probability that the deviation lies between -3.68 and 22.61, i.e., $P(-3.68 < X < 22.61)$.
   Standardize: $t_L = \frac{-3.68-5}{10} = -0.868$ and $t_U = \frac{22.61-5}{10} = 1.761$. Row 14: $P(T < -0.868) = 0.20$, $P(T > 1.761) = 0.05$. Thus $P(-0.868 < T < 1.761) = 1 - 0.20 - 0.05 =$ 0.75.

4. **Direct combined tails – revenue alerts.** The revenue deviation from plan over 6 weeks, in thousands of dollars, has mean 0 and approximate standard deviation 2. Model $X = 2T$, where $T \sim t_5$. An alert occurs when the deviation is below -5.142 or above 2.952. Find the alert probability, i.e., $P(X < -5.142 \text{ or } X > 2.952)$.
   Standardize: $t_L = \frac{-5.142-0}{2} = -2.571$ and $t_U = \frac{2.952-0}{2} = 1.476$. Row 5 gives lower tail 0.025 and upper tail 0.10. Disjoint tails add: 0.125, or 12.5%.

5. **Inverse left tail – temperature error.** The reading deviation from reference of 10 sensors, in degrees Celsius, has mean 2 and approximate standard deviation 4. Model $X = 2 + 4T$, where $T \sim t_9$. Find the cutoff x below which 10% of deviations fall, i.e., $P(X < x) = 0.10$.
   Row 9, $\alpha = 0.10$: $t = -1.383$. Hence $x = 2 + 4(-1.383) \approx$ −3.53 °C. About 10% of errors fall below this cutoff.

6. **Inverse right tail – processing error.** The processing-time deviation from target of 18 batches, in minutes, has mean -1 and approximate standard deviation 3. Model $X = -1 + 3T$, where $T \sim t_{17}$. Find the cutoff x exceeded by 5% of deviations, i.e., $P(X > x) = 0.05$.
   Row 17, $\alpha = 0.05$: $t = 1.740$. Hence $x = -1 + 3(1.740) \approx$ 4.22 minutes.

7. **Inverse interval – demand error.** The demand deviation from forecast over 14 weeks, in tonnes, has mean 5 and approximate standard deviation 10. Model $X = 5 + 10T$, where $T \sim t_{13}$. Find limits $L < U$ enclosing the central 90% of deviations, i.e., $P(L < X < U) = 0.90$, with equal excluded tails.

Each tail is 0.05, so row 13 gives $t^* = 1.771$. Thus $(L, U) = (5-10(1.771), 5+10(1.771)) \approx$ $(-12.71, 22.71)$ tonnes.

8. **Inverse combined tails – filling alerts.** The fill-mass deviation from target of 5 containers, in grams, has mean 0 and approximate standard deviation 2. Model $X = 2T$, where $T \sim t_4$. Find limits $L < U$ leaving 5% of deviations outside, equally split between the tails, i.e., $P(X < L) = P(X > U) = 0.025$.

   Each tail is 0.025. Row 4: $t^* = 2.776$. Therefore $(L, U) = (-2(2.776), 2(2.776)) \approx$ $(-5.55, 5.55)$ g; 2.5% lie in each tail.

PROBLEMS

Instructions. Use only the supplied Student's t critical-value table. Every specified standardized boundary is a rounded table entry. For direct problems, standardize and identify the appropriate tail probability; for inverse problems, find the table critical value and back-transform. Show your calculations and interpret results in context. No CDF or inverse-CDF calculator is needed. Show working on separate paper.

1. **Bus arrivals.** The arrival-time deviation from schedule of 7 buses, in minutes, has mean 2 and approximate standard deviation 6. Model $X = 2 + 6T$, where $T \sim t_6$. Find the probability that the deviation is below -9.658, i.e., $P(X < -9.658)$, and interpret the threshold.

2. **Shipping costs.** The cost deviation of 16 shipments, in thousand USD, has mean -15 and approximate standard deviation 40. Model $X = -15 + 40T$, where $T \sim t_{15}$. Find the probability that the deviation exceeds -49.64, i.e., $P(X > -49.64)$, and explain why it exceeds 0.5.

3. **Water demand.** The water-use deviation from forecast over 20 days, in cubic metres, has mean 10 and approximate standard deviation 20. Model $X = 10 + 20T$, where $T \sim t_{19}$. Find the probability that the deviation lies between 27.22 and 44.58, i.e., $P(27.22 < X < 44.58)$.

4. **Harvest forecasts.** The harvested-mass deviation from forecast of 9 harvests, in tonnes, has mean -2 and approximate standard deviation 4. Model $X = -2 + 4T$, where $T \sim t_8$. Find the probability that the deviation lies between -11.224 and -5.556, i.e., $P(-11.224 < X < -5.556)$.

5. **Weekly revenue alerts.** The revenue deviation from plan over 5 weeks, in thousands of dollars, has mean 1 and approximate standard deviation 3. Model $X = 1 + 3T$, where $T \sim t_4$. An alert occurs when the deviation is below -5.396 or above 5.599. Find the alert probability, i.e., $P(X < -5.396 \text{ or } X > 5.599)$.

6. **Cold-storage sensors.** The reading deviation from reference of 11 sensors, in degrees Celsius, has mean -2 and approximate standard deviation 6. Model $X = -2 + 6T$, where $T \sim t_{10}$. Find the cutoff x below which 10% of deviations fall, i.e., $P(X < x) = 0.10$.

7. **Rail journey delays.** The journey-time deviation from schedule of 24 rail journeys, in minutes, has mean 3 and approximate standard deviation 8. Model $X = 3 + 8T$, where $T \sim t_{23}$. Find the cutoff x exceeded by only 2.5% of journeys, i.e., $P(X > x) = 0.025$.

8. **Electricity forecasting.** The electricity-use deviation from forecast over 19 weeks, in kWh, has mean 10 and approximate standard deviation 40. Model $X = 10 + 40T$, where $T \sim t_{18}$. Find limits $L < U$ enclosing the central 90% of deviations, i.e., $P(L < X < U) = 0.90$, with equal excluded tails.

9. **Packaging inspection.** The mass deviation from target of 6 packages, in grams, has mean -0.5 and approximate standard deviation 1.5. Model $X = -0.5 + 1.5T$, where $T \sim t_5$. Find limits $L < U$ leaving 5% of deviations outside, equally split between the tails, i.e., $P(X < L) = P(X > U) = 0.025$.

10. **Processing-time alerts.** The processing-time deviation from target of 13 batches, in minutes, has mean 1 and approximate standard deviation 5. Model $X = 1 + 5T$, where $T \sim t_{12}$. Find limits $L < U$ leaving 15% of deviations outside, with 5% below L and 10% above U, i.e., $P(X < L) = 0.05$ and $P(X > U) = 0.10$.

Detailed Solutions

All critical values are read from the supplied table and rounded to three decimals. Probabilities are table-implied; transformed cutoffs are approximate.

1. **Bus arrivals ($\nu = 6$).**
   $t = \frac{-9.658-2}{6} = -1.943$. Row 6, $\alpha = 0.05$:
   $$P(T < -1.943) = \boxed{0.05}.$$
   The cutoff is approximately -9.66 minutes; about 5% arrive more than 9.66 minutes early.

2. **Shipping costs ($\nu = 15$).**
   $t = \frac{-49.64+15}{40} = -0.866$. Row 15, $\alpha = 0.20$:
   $$P(T > -0.866) = 1 - 0.20 = \boxed{0.80}.$$
   The threshold is -49.64 thousand USD relative to budget. It is below the location -15 thousand USD, so more than half of outcomes exceed it.

3. **Water demand ($\nu = 19$).**
   $t_L = \frac{27.22-10}{20} = 0.861$, $t_U = \frac{44.58-10}{20} = 1.729$. Row 19: $P(T > 0.861) = 0.20$ and $P(T > 1.729) = 0.05$.
   $$P(0.861 < T < 1.729) = 0.20 - 0.05 = \boxed{0.15}.$$
   The bounds are 27.22 and 44.58 cubic metres above forecast; about 15% of days fall between them.

4. **Harvest forecasts ($\nu = 8$).**
   $t_L = \frac{-11.224+2}{4} = -2.306$, $t_U = \frac{-5.556+2}{4} = -0.889$. Row 8 gives $P(T < -2.306) = 0.025$ and $P(T < -0.889) = 0.20$.
   $$P(-2.306 < T < -0.889) = 0.20 - 0.025 = \boxed{0.175}.$$
   The interval is approximately $(-11.22, -5.56)$ tonnes; about 17.5% of harvests fall short by this amount.

5. **Revenue alerts ($\nu = 4$).**
   $t_L = \frac{-5.396-1}{3} = -2.132$, $t_U = \frac{5.599-1}{3} = 1.533$. Row 4: lower tail 0.05, upper tail 0.10.
   $$P(\text{alert}) = 0.05 + 0.10 = \boxed{0.15}.$$
   The alerts occur below -5.40 or above 5.60 thousand dollars relative to plan. About 15% of weeks trigger an alert.

6. **Cold-storage sensors ($\nu = 10$).**
   Lower-tail probability 0.10 corresponds to $t = -1.372$ (row 10).
   $$x = -2 + 6(-1.372) \approx \boxed{-10.23\ ^\circ\text{C}}.$$
   About 10% of sensor errors fall below -10.23 °C relative to reference.

7. **Rail journey delays ($\nu = 23$).**
   Upper-tail probability 0.025 corresponds to $t = 2.069$ (row 23).
   $$x = 3 + 8(2.069) \approx \boxed{19.55 \text{ minutes}}.$$
   About 2.5% of journeys are more than 19.55 minutes late.

8. **Electricity forecasting ($\nu = 18$).**
   Central 90% leaves 0.05 in each tail. Row 18, $\alpha = 0.05$: $t^* = 1.734$.
   $$L = 10 - 40(1.734) \approx \boxed{-59.36 \text{ kWh}},$$
   $$U = 10 + 40(1.734) \approx \boxed{79.36 \text{ kWh}}.$$
   About 90% of errors lie inside; 5% lie in each tail.

9. **Packaging inspection ($\nu = 5$).**
   Each excluded tail has probability 0.025. Row 5: $t^* = 2.571$.
   $$L = -0.5 - 1.5(2.571) \approx \boxed{-4.36 \text{ g}},$$
   $$U = -0.5 + 1.5(2.571) \approx \boxed{3.36 \text{ g}}.$$
   About 95% of mass errors lie between the limits, with 2.5% on each side.

10. **Processing-time alerts ($\nu = 12$).**
    Lower tail 0.05 gives $t_L = -1.782$; upper tail 0.10 gives $t_U = 1.356$ (row 12).
    $$L = 1 + 5(-1.782) \approx \boxed{-7.91 \text{ minutes}},$$
    $$U = 1 + 5(1.356) = \boxed{7.78 \text{ minutes}}.$$
    About 5% of outcomes lie below L, 10% above U, and 85% between them.